PRODUCT MANAGEMENT WITH GENERATIVE & AGENTIC AI





Problem Statement

How can the process of travel planning be simplified by designing a multi-agent system that can handle research, bookings, budget optimisation, and contextual recommendations?

Team Details & Contribution

Team Members	Student Ids	Contribution
Kirti	BITSoM_PM_2511852	Product Ideation, Framework Designing & Brainstorming, Value Proposition Canvas, Determining the unit of sales, Problem signal & reviews from different platforms.
Syed Mairaj Mohiuddin	BITSoM_PM_2511846	Product Ideation, Framework Designing & Brainstorming Revenue Models, How to forecast the demand, Problem signal & reviews from different platforms.
Jigeesha Jain	BITSoM_PM_2511850	Product Ideation, Framework Designing & Brainstorming Who is our competition, Why will users switch to our solution, Problem signal & reviews from different platforms.
Biswa Bhusan Dash	BITSoM_PM_2511854	Product Ideation, Framework Designing & Brainstorming What are the key metrics, Key elements of the cost structure, Problem signal & reviews from different platforms.
Kandarp Sojitra	BITSoM_PM_2511834	Product Ideation, Framework Designing & Brainstorming Who are the customers, CVP creation, Problem signal & reviews from different platforms.

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"Mosaic Trip" - From inspiration to execution, without the planning chaos

1.Exploring Product Ideas

The core idea is to design a **Multi-Agent Travel Planning System** where specialized AI agents collaborate to simplify end-to-end travel planning. Instead of users manually switching between multiple apps and websites, the system works as a single intelligent assistant.

Proposed Agents & Responsibilities

- **Research Agent**
  Collects and summarizes information on destinations, best travel seasons, visa rules, weather, attractions, and local experiences.
- **Booking Agent**
  Handles flight, hotel, and activity bookings by comparing options across platforms and suggesting the best choices based on user preferences.
- **Budget Optimisation Agent**
  Allocates budget intelligently across flights, stays, food, and experiences while suggesting cost-saving alternatives.
- **Contextual Recommendation Agent**
  Provides personalized suggestions based on travel purpose (leisure/work), companions, weather, local events, and user history.

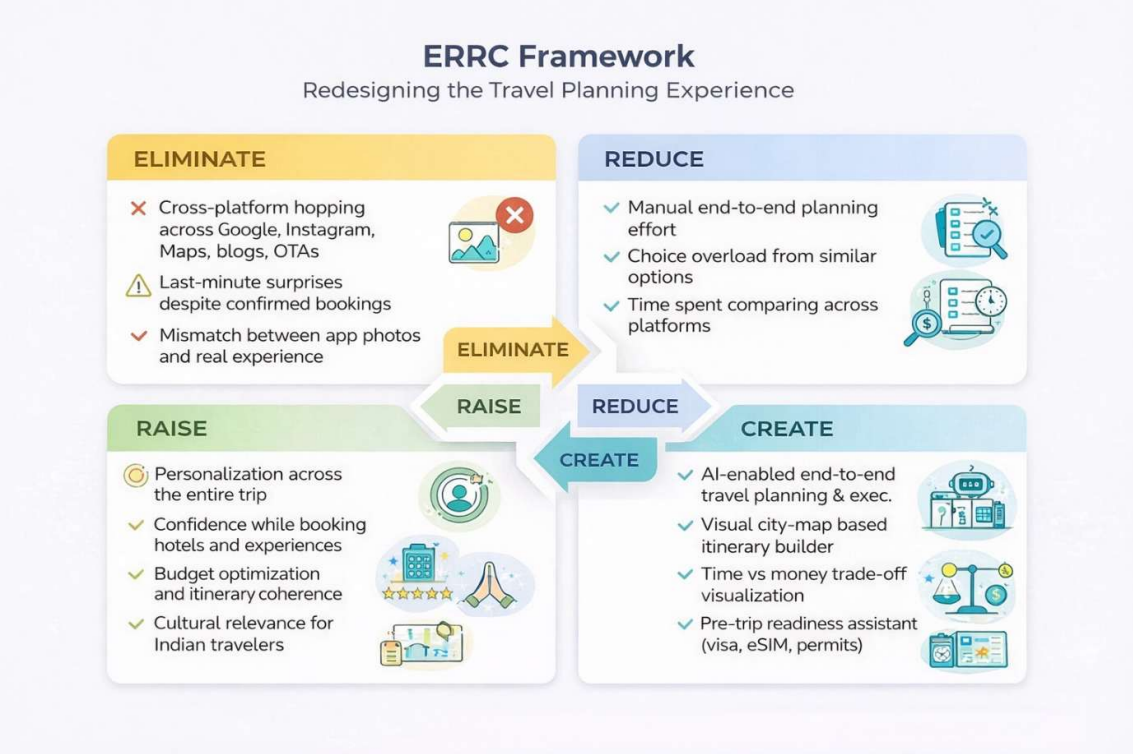
```
User → Orchestrator Agent
              ↓
| Research Agent | Booking Agent | Budget Agent | Context Agent |
```

Each agent has **clear responsibility, inputs, outputs, and KPIs**.



2. Frameworks:

a. ERRC



b. SCAMPER



c. Brainstorming Outcomes

Collaborative brainstorming sessions led to the following high-impact ideas:

- A **single-chat travel planning experience** from intent to booking
- Smart **budget distribution recommendations** across flights, stays, and experiences
- **Auto-replanning** in case of delays, cancellations, or budget changes
- Context-aware **day-wise itineraries** instead of rigid plans
- Seamless **group travel collaboration** with shared preferences and budgets

3. Sources for These Ideas

The product ideas were derived from a mix of qualitative insights, competitive research, and real-world observations.

- **User Pain Points**: Difficulty in comparison, overwhelming choices, budget overruns, and lack of personalization

- **Competitive Analysis**: Evaluation of platforms like MakeMyTrip, Booking.com, Google Travel, and TripAdvisor
- **Personal Experiences**: Challenges faced during multi-city, budget-constrained, and group travel planning
- **Industry Trends**: Growth of AI agents, automation, and hyper-personalization
- **Secondary Research**: Travel blogs, UX case studies, online forums, and expert articles

4. How Will the Ideas Be Gathered?

A structured and user-centric research approach will be followed to continuously refine ideas.

- **User Interviews**: Discussions with frequent travellers, solo travellers, and first-time planners
- **Surveys & Questionnaires**: To identify key frustrations, expectations, and success criteria
- **Competitive Feature Mapping**: Identifying gaps and opportunities in existing platforms
- **Online Reviews & Forums**: Insights from Reddit, TripAdvisor, and app store reviews
- **Iterative Feedback Loops**: Rapid prototyping followed by usability testing and refinement

5. Who are our customers?

The following segments represent the core audience and key partners for our offerings:

Customer Segment	Description	Key Needs & Characteristics
Young Travelers & Weekend Explorers	Individuals/small groups seeking short, easily accessible, and budget-friendly trips.	Spontaneity, convenience, cost-effectiveness, unique experiences, digital booking/support.
Family Vacation Travelers	Families (with children of various ages) planning multi-day holidays requiring comfort, safety,	Safety, reliability, structured itineraries, comfortable accommodation, value for

	and activities suitable for all members.	money, child-friendly options.
Pilgrimage & Cultural Travelers	Individuals and groups focused on spiritual journeys, temple visits, and deep immersion in local culture and heritage.	Authentic experiences, proximity to religious sites, specialized services and divine experience
Adventure & Trekking Enthusiasts	Customers seeking physically demanding activities, nature exploration, and off-the-beaten-path destinations.	Specialized equipment/gear support, experienced guides, robust safety protocols, detailed route planning.

6. Who is our competition?

Category	Description	Examples
Online Travel Agencies (OTAs)	Primary competitors that handle bookings and basic trip planning. Users still need to manually research, compare options, and coordinate multiple aspects of travel.	MakeMyTrip (MMT), Booking.com, Goibibo, Agoda
Itinerary Planning & Travel Inspiration Tools	Platforms focused on itinerary creation and travel inspiration, but lacking full execution and booking capabilities.	TripIt, Wanderlog, TripAdvisor
AI-Based Travel Assistants & Chatbots	Emerging AI-driven solutions that provide conversational planning and recommendations, with limited or partial booking support.	ChatGPT (travel planning), Google Gemini, Expedia Group (Romie), Navan

Traditional Travel Agents & Corporate Travel Desks	Human-led services offering personalized travel planning, but constrained by scalability, speed, and real-time adaptability.	Offline travel agencies, corporate travel management companies

7. Customer Value Proposition

For **travellers tired of juggling apps and last-minute surprises**, **Mossaic Trip** our AI-powered planner helps travellers to plan **effortless, personalized trips saving time**, cutting confusion, and locking in real experiences. Unlike typical travel apps, we craft **smart, budget-friendly itineraries tailored for personalized choices**.

Segment Specific CVP

Young Travelers: Instant, budget-friendly itineraries for stress-free, authentic weekend adventures.

Family Travelers: Effortless planning for safe, customized family holidays with guaranteed comfort and lasting memories.

Pilgrimage & Cultural Travelers: We Provide Divine, stress-free spiritual journeys guided by local expertise, ensuring proximity to sacred sites and deep cultural immersion.

8. Value Proposition Canvas



9. Reasons why Customers would switch to our Product

1) One Platform for Everything

Imagine a single platform where research agents, booking agents, budget optimizers, and recommendation engines all work seamlessly together—automatically. Users no longer need to waste time manually comparing options; everything is intelligently coordinated for them.

2) Real-Time Budget Optimization

Unlike traditional, static travel apps, our system actively searches and finds the perfect mix of flights, hotels, experiences, and transportation — all tailored dynamically to fit each user's budget in real time.

3) Zero Manual Effort

Planning travel has never been easier. Customers just share their goals, and our system takes over:

1. Researching
2. Filtering
3. Comparing
4. Crafting the ideal itinerary

This process saves users hours of tedious planning and eliminates decision fatigue.

Customers will switch to us as we would transform travel planning from a frustrating chore into a smooth, enjoyable experience. Instead of juggling multiple websites to compare prices, read reviews, track budgets, and book separately, our intelligent multi-agent system does it all — smartly and automatically.

What users gain:

- Time savings: We handle research and booking for them
- Better decisions: Powered by constant price scanning and budget optimization
- Peace of mind: Every element of the itinerary is aligned, cross-checked, and optimized

Travel smarter, not harder.

10. Units of Sale

1. Complete Trip (End-to-End Journey)
Primary & Strongest Unit
A fully planned and executed trip covering all stages: research → booking → budgeting → recommendations.
Fully planned itinerary on the Mosaic Tour App
2. Booking Transactions (The Core Unit)

This is the most common unit, where the platform earns a commission or a markup on an individual reservation.
Per Air Ticket: Commission or convenience fees per passenger/segment.
Per Room Night: For hotels and homestays, the unit is typically one room for one night.
Per Holiday Package: A bundled unit (flight + hotel + transfers) sold as a single "tour."
Per Ticket/Seat: For bus, rail, or local activities/attractions.

3. Service & Convenience Fees
These are secondary units applied to the core transaction to cover the cost of the platform's technology and customer service.
Convenience Fee: A flat fee per booking (usually applied to flights).
Ancillary Services: Units like travel insurance, "zero-cancellation" protection, or premium seat selection.

3. Subscription & Loyalty Units
Mosaic Trip can use recurring models to lock in users and increase "Life Time Value" (LTV).
Black/Double Black Membership: A flat fee for a time-bound subscription (e.g., annual or semi-annual) that offers benefits like free cancellations or lounge access.

4. B2B & Advertising Units
As we onboard users, we can also sell "real estate" on the app to other brands.
Cost Per Click (CPC) / Cost Per Impression (CPM): Selling "Sponsored" or "Promoted" slots to hotels or airlines to appear higher in search results.
Leads: Charging boutique travel agencies for customer leads for specialized tours.

11. Revenue Models

1. The Transaction Revenue Model
This is our primary driver for **Flights** and **Budget Hotels**.

- **How it works:** The travel management App acts as a marketplace. The user pays the service provider through the app, and the app keeps a percentage.
- **Margin: 5% – 8%** for flights; **12% – 15%** for domestic hotels.
- **Strategic Role:** High-frequency use. It brings users to the app (Acquisition).

2. The Ancillary Revenue Model
Ancillaries are the fastest-growing segment (growing at ~45% YoY). These are "add-ons" that solve travel pain points.

- **Travel Insurance:** Partner with providers (like ACKO or Digit) to offer one-click insurance. A **20% – 30% commission** on the premium can be settled with the partners.
- **Visa Processing:** Visa assistance is a useful service that customers can avail.
- **SIM Cards:** Integrated "Travel Kits", SIM services for outbound travellers.
- **Convenience Fees:** Charging a flat fee (e.g., ₹200–₹500) per booking for "platform maintenance."

3. The Ad-Tech & Data Model

Once the product hits a significant user base, user data becomes a product itself.

- **Sponsored Listings:** Hotels pay a "bid" (Cost-Per-Click) to appear at the top of search results (similar to Amazon or TripAdvisor).
- **Destination Marketing (B2G):** Partnering with tourism boards (e.g., Kerala Tourism or Thailand Tourism) to run targeted campaigns to your high-intent users.
- **Brand Partnerships:** Displaying ads for travel-related goods (luggage, cameras, outdoor gear) during the "Planning Phase."

4. Subscription & Loyalty Model

To increase annual spend, the App needs users to book with exclusivity.

- **Tiered Memberships:** A "Pro" or "Gold" tier where users pay an annual fee (e.g., ₹999/year) in exchange for:
    - Zero convenience fees.
    - Priority customer support.
    - Early access to "Flash Sales"

5. Spiritual Experience Marketplace

Transactional Commission (The "Darshan" Fee)

- **Mechanism:** Partner with temple boards (like TTD, Vaishno Devi Shrine Board) to offer slot bookings directly in the app.
- **Take Rate: 8% – 12%** per booking.

12. Demand Forecasting

For calculating the Demand forecast, initially the TAM, SAM, and SOM of the Indian Travel Industry has been estimated and validated using both Top Down and Bottom Up Approaches as below.

Using the SOM arrived at, we would be forecasting the Revenue/Volume Demand

Calculation of TAM, SAM, and SOM

Top-Down Approach				
		Value(USD)	**Value(INR)**	**Source**
Total Addressable Market (TAM)	**Indian Travel Market Size estimate for FY26(Domestic and Outbound**	**250 Bn**	**21 Tn**	**VIDEC India Travel Market Report https://videc.co/wp-content/uploads/2024/02/VIDECs-India-Travel-Market-Opportunity-FY20-FY26-**

	bookings made by Indian travellers)			**Market-Sizing-OTA-Benchmarking-Modules.pdf**
Serviceable Addressable Market (SAM)	**Indian Online Travel Market Size in FY24**	**50 Bn**	**4.5 Tn**	**India Online Travel Market Size, Share, Trends, and Forecast https://www.imarcgroup.com/india-online-travel-market**
Serviceable Obtainable market (SOM)	**Major OTAs' market share (FY24) MakeMyTrip - 55.3% Clear Trip - 8.7% ixigo Group - 8.6% For a new entrant, aiming for a conservative 2% market share of the online segment within the first 6 years of operations**	**1 Bn**	**0.09 Tn**	**https://videc.co/makemytrip-emerges-as-dominant-player-in-indian/**

Bottom-Up Approach		
	Value (USD)	**Source**
Total Online travel booking population(estd.) - FY 25 figures	150 Mn	https://www.partners.skyscanner.net/news-case-studies/unlocking-indias-rapidly-expanding-travel-economy
Average Annual Travel Expenditure Millennials - $6,031 Gen Z - $2,622 Gen X - $3,059	378	https://www.business-standard.com/india-news/india-s-millennials-spend-6-031-on-travel-leading-gen-z-gen-x-boomers-124091200783_1.html https://videc.co/wp-content/uploads/2024/02/VIDECs-

Boomers - $2,600 Blended average as per the generational composition - $3,781.47 OTA capture ratio - 40% Considering a conservative 10% capture ratio in the case of a new entrant: Average Final Travel Expenditure: $378 Approx		India-Travel-Market-Opportunity-FY20-FY26-Market-Sizing-OTA-Benchmarking-Modules.pdf
Total Serviceable Addressable Market - SAM	56.7 Bn	

Now that we have the SAM validated, we can proceed with the SOM of $ 1 Bn as an achievable target for the product within a span of 6 years.

Revenue & Volume Forecast

Assumptions	**Value**
Average Annual Spend	**378$**
Take Rate of OTAs	**10.50%**
SOM (within 6 years)	**$1000 Million**

Volume Forecast			
Year	**Target Gross Booking Value (Million USD)**	**Required Active Transacting Users (at 378$ of average annual spend)**	**Active Users (Lac)**
1	50	132,275	1.3
2	75	198,413	1.9
3	150	396,825	3.9
4	300	793,651	7.9
5	600	1,587,302	15.8
6	1000	2,645,503	26.5

Revenue Forecast			
Year	**Target Gross Booking Value (Million USD)**	**Net Revenue projected at a take rate of 10% (Million USD)**	**Net revenue (INR)**
1	50	5	455,000,000
2	75	7.5	682,500,000
3	150	15	1,365,000,000
4	300	30	2,730,000,000
5	600	60	5,460,000,000
6	1000	100	9,100,000,000

13. Key elements of the cost structure

The cost structure of the proposed AI-enabled travel planning and execution platform is driven by the need to deliver **personalized, reliable, and end-to-end trip experiences** rather than just bookings. The key elements are as follows:

1. Technology & Platform Infrastructure

- Cloud hosting and scalable backend infrastructure to support search, planning, bookings, and real-time updates
- Development and maintenance of web and mobile applications
- Integration costs for third-party APIs such as flights, hotels, maps, routing, and payments

Rationale:
Technology is a core enabler of the product's value proposition and must scale reliably during peak travel periods.

2. AI & Personalization Engine

- Development, training, and ongoing improvement of AI models for itinerary creation, recommendations, and trade-off simulations
- Costs associated with running AI models in real time (inference)
- Experimentation and A/B testing to improve recommendation quality

Rationale:
Hyper-personalization and planning intelligence are key differentiators and require continuous investment.

3. Data & Content Acquisition

- Access to high-quality third-party data (hotels, activities, transport, reviews)
- Verification of listings, images, and location accuracy to reduce experience mismatch
- Local, cultural, and India-specific data to support family-friendly and religious travel needs

Rationale:
Trust and confidence in planning depend on accurate, reliable, and context-aware data.

4. Customer Acquisition & Marketing

- Performance marketing and app-install campaigns
- Content and influencer marketing for travel inspiration
- Referral programs and loyalty incentives to drive repeat usage

Rationale:
The travel category is competitive, and sustained user acquisition is essential for scale.

5. Transaction & Payment Processing

- Payment gateway and settlement fees
- Refund, cancellation, and rescheduling costs
- Fraud prevention and chargeback handling.

Rationale:
End-to-end booking and post-booking support increase transaction-related operational costs.

6. Customer Support & Operations

- AI-assisted and human customer support for pre-trip, on-trip, and post-trip issues
- Handling last-minute changes, disruptions, and emergencies
- Partner coordination during execution

Rationale:
Execution support is a key point of differentiation from existing OTAs and requires ongoing operational investment.

7. Partner Onboarding & Management

- Onboarding and managing hotels, transport providers, and local experience partners
- Quality checks, compliance verification, and partner relationship management

Rationale:
A trusted partner ecosystem reduces customer pain but adds operational overhead.

8. Compliance, Security & Risk Management

- Data privacy and security compliance costs
- Regulatory and legal costs related to travel and payments
- Risk mitigation and insurance where applicable

Rationale:
Trust, safety, and compliance are critical in travel, especially for families and group travelers.

14. Key Metrics

The success of the AI-enabled travel planning and execution platform will be measured using metrics that track **user effort reduction, decision confidence, personalization quality, execution reliability, and business viability**.

1. User Effort & Planning Efficiency

These metrics assess whether the product reduces planning chaos and cognitive load.

- **Time to create an itinerary** (hours/minutes)
- **Number of steps or inputs required** to finalize a trip plan
- **Drop-off rate during planning flow**

Why it matters:
Directly measures the “reduce effort” and “eliminate cross-platform hopping” goals.

2. Decision Quality & Confidence

These metrics capture how confidently users make booking decisions.

- **Time to final decision** (hotel/flight/activity)
- **Booking completion rate**
- **Pre-booking confidence score** (quick in-app survey)
- **Reduction in comparison loops**

Why it matters:
Indicates whether the product reduces confusion and decision fatigue.

3. Personalization & Recommendation Effectiveness

These metrics evaluate how well the AI recommendations fit user preferences.

Recommendation acceptance rate

- **"This fits my preferences" CSAT score**
- **Repeat usage of recommended items**
- **Saved preference reuse across trips**

Why it matters:
Personalization is a key differentiator versus existing OTAs.

4. Execution & Reliability Metrics

These metrics measure how smoothly the trip runs after booking.

- **Booking success rate**
- **Cancellations or changes after confirmation (%)**
- **On-ground issue rate** (support tickets during trip)
- **Alerts triggered vs issues prevented**

Why it matters:
Shifts focus from booking completion to actual trip success.

5. Trust & Satisfaction Metrics

These metrics assess long-term trust and user delight.

- **Net Promoter Score (NPS)**
- **Post-trip CSAT**
- **App store ratings and reviews**
- **Post-booking regret score**

Why it matters:
Trust and satisfaction drive repeat usage in travel products.

6. Business & Growth Metrics

These metrics ensure the product is commercially viable.

Monthly Active Users (MAU)

- **Repeat booking rate**
- **Average order value (AOV)**
- **Revenue per user**
- **Customer acquisition cost (CAC) vs lifetime value (LTV)**

Why it matters:
Connects product success to sustainable business outcomes.

15. Experimental Task

Problems that Users face with Online Travel booking apps and reviews posted online

As part of the experimental task, we went through user reviews that were posted for different travel management apps like MakeMyTrip, Goibibo, on platforms like Reddit, X, LinkedIn, and Google Reviews

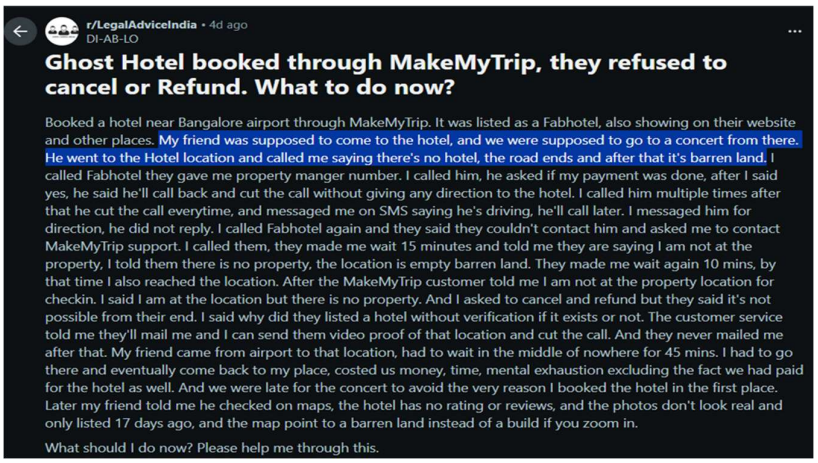
Below are the major problems users faced with the particular social media post links of the user reviews

1. **Issues with Hotel Bookings**

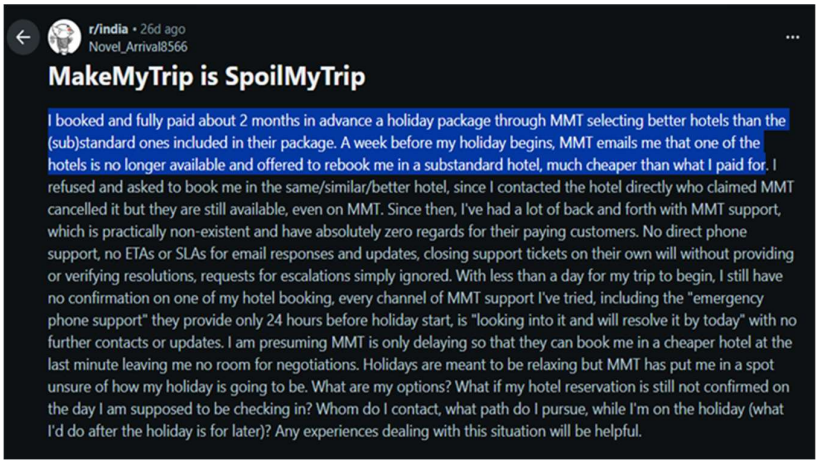
Users complain of the following issues with respect to the Hotels that they book on the travel Apps
a) Ghost properties – A hotel that is booked on the App is not present at the location
b) Bad experience with the hotels – Hotel Image,s which are posted on the Apps, are completely misleading, and when the users reach the exact location, they end up finding bad quality service from the Hotels

User reviews with Links

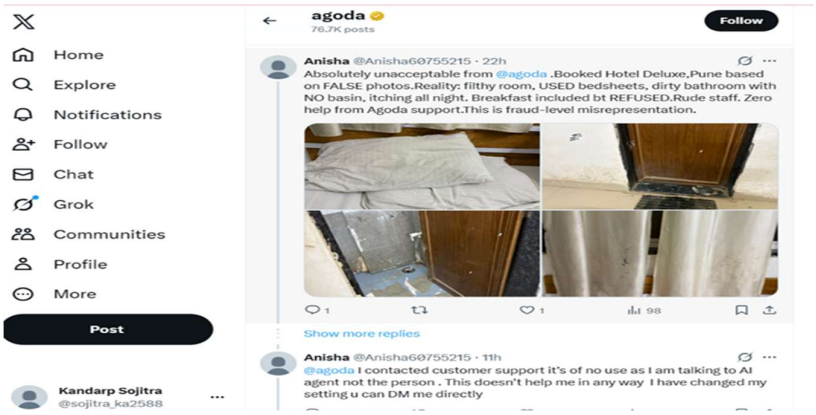
[User complaining of a Missing Hotel on Reddit]



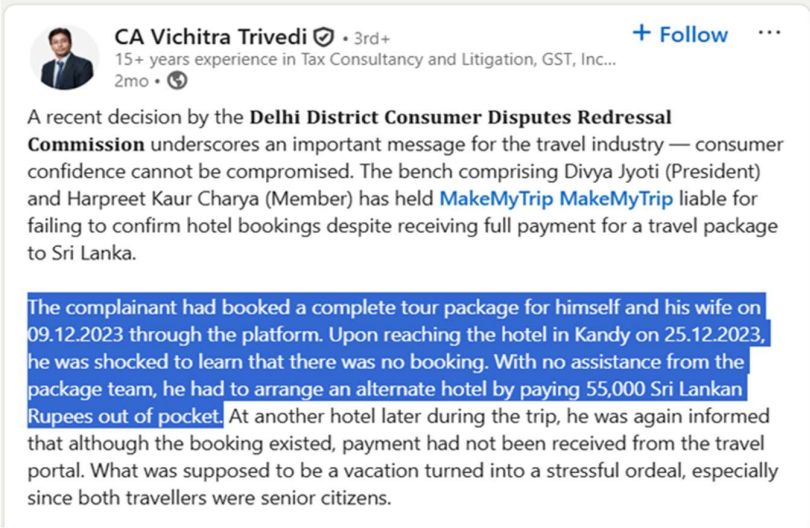
[User complaining on Reddit about Last Minute ‘Hotel Not Available’ Downgrade, wherein the hotel was booked well in advance]



User complaining on X about a bad hotel experience in a Hotel booked through Agoda



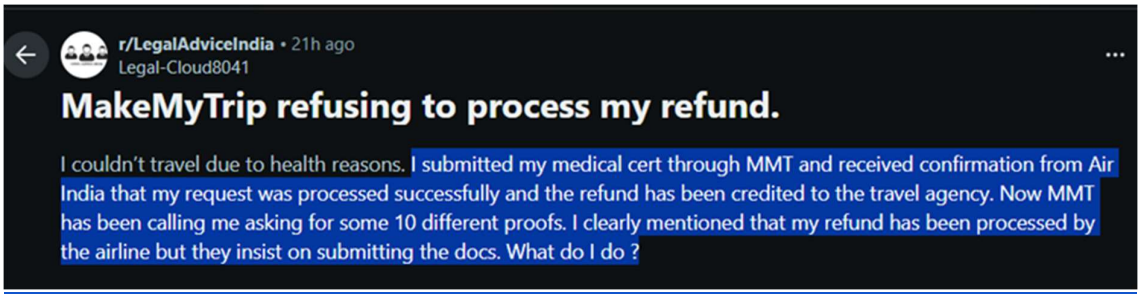
User complaining on LinkedIn about last minute denial of a hotel booking, which was booked through MakeMyTrip Well in advance



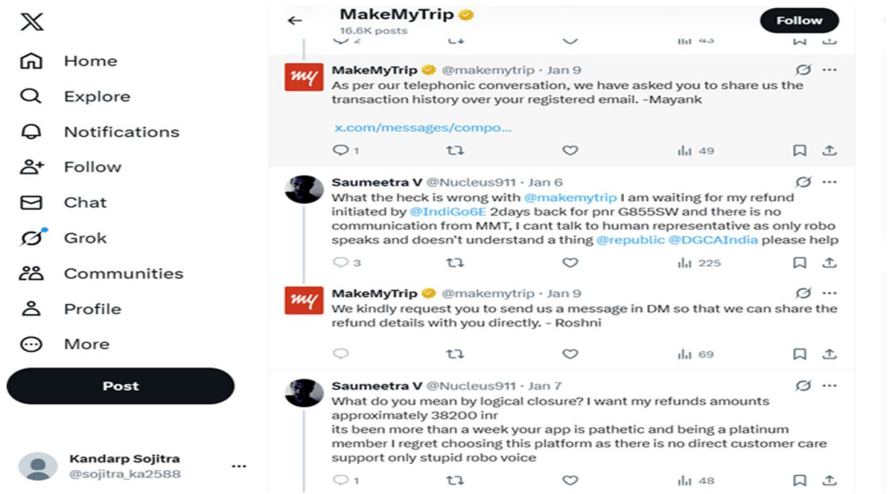
2. Refund Issues

Users complain with respect to delayed refunds, no response from the Online travel management apps regarding the refund queries, and in some cases, Apps making it very difficult for users to get their refunds processed.

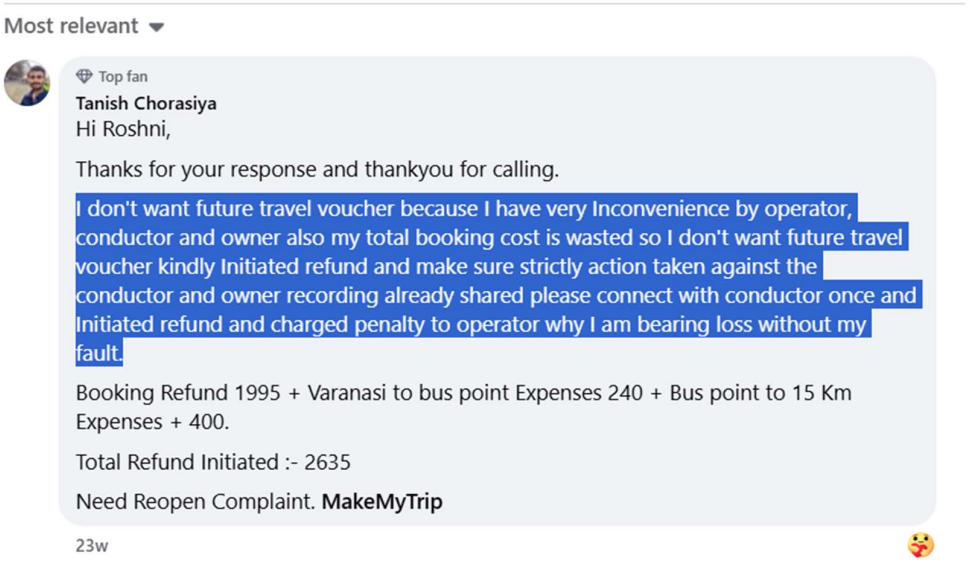
User complaining about MakeMyTrip on Reddit about delayed refunds



User complaining about delayed refunds from MakeMyTrip and improper response on X



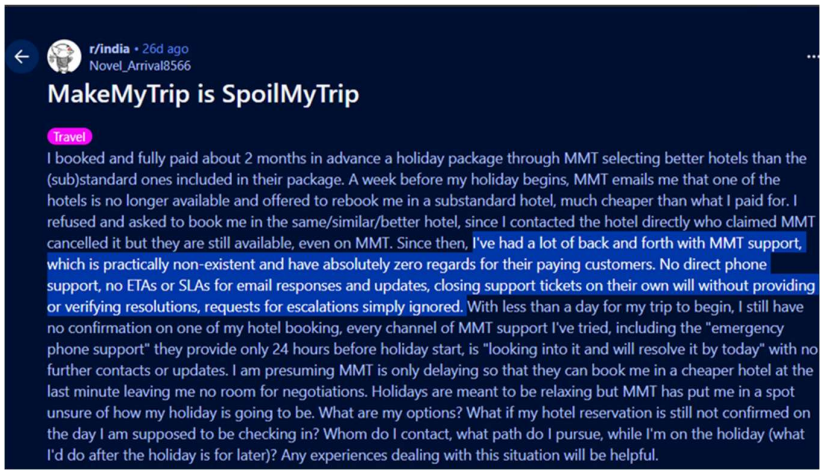
User complaining to MMT support on Facebook about travel vouchers offered as part of refund



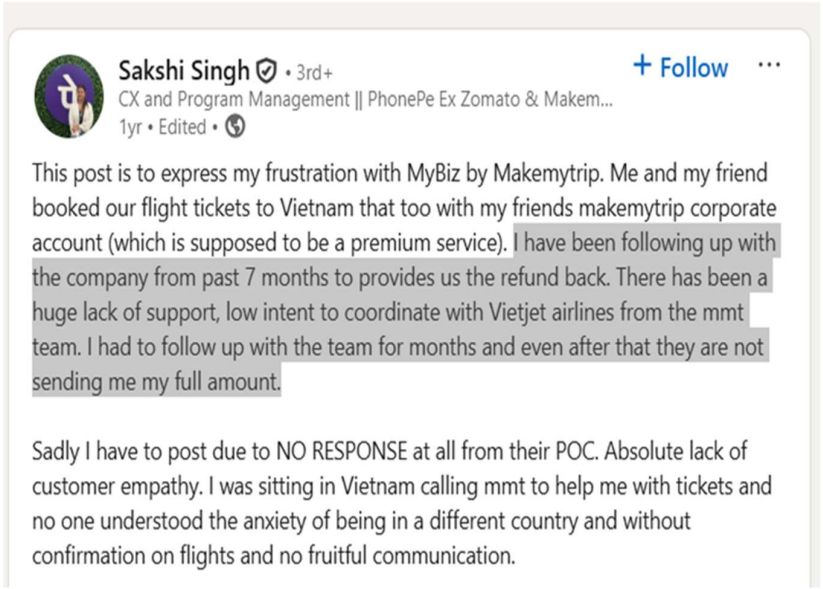
3. Improper Customer Support

Users complain about not receiving proper Customer support from the prominent OTAs and an AI bot loop that keeps giving them repeated generic replies.

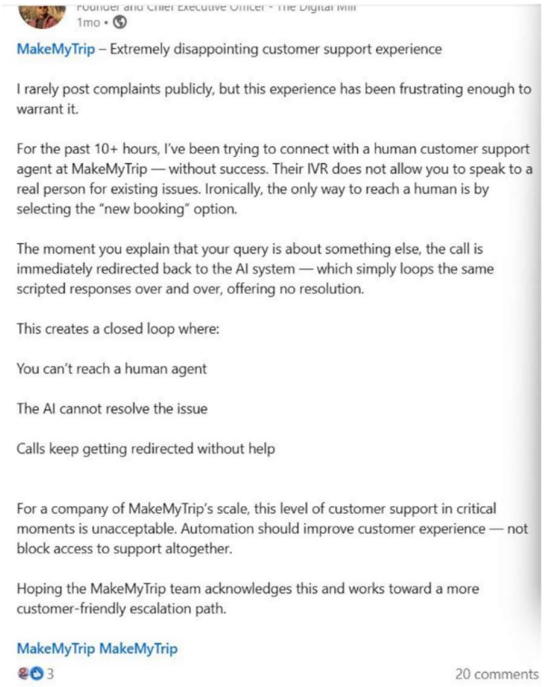
User complaining on Reddit about improper customer support from MakeMyTrip



User complaining on LinkedIn about the lack of response from the MakeMyTrip customer support



[User complaining about MakeMyTrip's improper customer support on LinkedIn]



Founder and Chief Executive Officer - The Digital Mill
1mo •

MakeMyTrip – Extremely disappointing customer support experience

I rarely post complaints publicly, but this experience has been frustrating enough to warrant it.

For the past 10+ hours, I've been trying to connect with a human customer support agent at MakeMyTrip — without success. Their IVR does not allow you to speak to a real person for existing issues. Ironically, the only way to reach a human is by selecting the "new booking" option.

The moment you explain that your query is about something else, the call is immediately redirected back to the AI system — which simply loops the same scripted responses over and over, offering no resolution.

This creates a closed loop where:

You can't reach a human agent

The AI cannot resolve the issue

Calls keep getting redirected without help

For a company of MakeMyTrip's scale, this level of customer support in critical moments is unacceptable. Automation should improve customer experience — not block access to support altogether.

Hoping the MakeMyTrip team acknowledges this and works toward a more customer-friendly escalation path.

MakeMyTrip MakeMyTrip

3 20 comments